



Diffusion in confinement as a microscopic relaxation mechanism in glass-forming liquids

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ABSTRACT

Using quasielastic neutron scattering, we compare dynamics in single-element liquids, glass-forming selenium and non glass-forming gallium. There is a single jump-diffusion process in gallium, whereas in selenium there is also a faster, spatially localized process. The fast and slow processes describe β - and α -relaxation, respectively. We then analyze an archetypical glass-former, glycerol, to show that the two-component fit, with β - and α -relaxations explicitly separated, yields the correct value for the translational diffusion coefficient and provides information on the spatial localization of the β -relaxation that is not experimentally accessible otherwise.

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1. Introduction

Studies of glass-forming liquids tend to concentrate on the supercooled regime, where the main relaxation exhibits dramatic slowing down near the glass transition temperature, T_g . In this work, however, we investigate the behavior of glass-forming liquids above melting temperature, T_m , where the temperature dependence of the main relaxation is nearly Arrhenius, and no macroscopic measurement is able to predict eventual glass formation. However, microscopic techniques with sensitivity to the dynamic processes on variable length scales can identify in both modeling (molecular dynamics simulation, MD) and experiments (quasielastic neutron scattering, QENS) the features that may be predictive of the glass-forming capabilities of the system. In this work we use QENS to compare high-temperature ($T \approx 1.2T_m$) dynamics of a glass-forming (selenium) and a non-glass forming (gallium) liquid. As any liquid, both gallium and selenium exhibit long-range translational diffusion. Unlike its non glass-forming counterpart, selenium also exhibits a faster relaxation process, which is spatially localized. In the energy space, these two processes, long-range and localized, manifest themselves in the presence of two dynamic components in the dynamic structure factor, $S(Q, E)$, with a distinctive dependence on the momentum scattering transfer, Q . In the time space, this corresponds to a two-step decay of the intermediate scattering function, $I(Q, t)$. Such decay, first to a finite value through a secondary β -relaxation, then to zero through the main α -relaxation, is predicted by mode-coupling theory (MCT) [1–3]. It has been reported in various MD studies of glass-forming liquids [4–6], which, however, tend to analyze the $I(Q, t)$ at Q values corresponding to the first structural maximum of the

static structure factor, $S(Q)$. The same applies to the experimental studies of supercooled liquids [7,8]. Only recently there have been MD [9,10] and QENS [11–13] studies that present the signature of the two-step relaxation in the $I(Q, t)$ and the $S(Q, E)$ at low wave numbers, which is particularly relevant in systems that scatter neutrons predominantly incoherently and, thus, yield information on the single-particle dynamics.

At first, we compare single-element liquids, selenium and gallium, both of which exhibit propensity for supercooling below their respective melting temperatures without freezing. However, selenium readily forms glass at regular cooling rates, while gallium does not, and will spontaneously crystallize instead. Selenium was chosen as the only single-element glass-former that could be produced by viscous slowdown of the liquid at moderate cooling rates and ambient pressure [14]. Gallium was selected for its unusual, for a non glass-former, propensity for supercooling below T_m . Neither selenium nor gallium is a simple liquid. The melting temperature of solid gallium is lowered by pressure, and its tendency to form strong directed covalent bonds between pairs of atoms persists in both solid and liquid states. Selenium is believed to be a mixture of chains and eight-atom rings [15], although *ab initio* MD simulations show that the fraction of ring molecules is small [16–18]. Furthermore, the incoherent neutron scattering cross-sections of gallium and selenium are small: 0.16 barn and 0.32 barn, respectively, which makes an incoherent QENS experiment difficult. This may explain relative lack of QENS data on liquid gallium [19] and selenium [20–22]. Neutron spin-echo studies of liquid selenium [8] concentrated on the coherent scattering. Our goal was to choose specifically single-element liquids to see how the ‘caging’ effect by the neighboring particles manifests itself when there is no contribution to the ‘caging’ from chemical heterogeneity. Having observed ‘caging’, or β -relaxation, in a single-element liquid glass-former with no chemical heterogeneity, selenium, we

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then switch to an archetypical molecular liquid glass-formers, glycerol. Glycerol is hydrogen-bearing and thus scatters neutrons predominantly incoherently, conveniently lending itself to incoherent QENS measurements of the single-particle dynamics. Our data analysis in the framework of the confined single-particle diffusion describes the β -relaxation and also yields the correct value of the long-range translational diffusion coefficient (α -relaxation). Thus, the information on the spatial characteristics of the β -relaxation can be extracted from the low- Q QENS data, as opposed to, e.g., data collected near the structural maximum position. This kind of information is not accessible by other techniques, except in MD simulations.

2. Experiment

QENS experiments were performed at the Spallation Neutron Source, Oak Ridge National Laboratory (ORNL) using backscattering spectrometer BASIS [23] in the standard operation regime, with 3.4 μeV energy resolution (full-width at half-maximum, Q -averaged value) and the dynamic range of $\pm 100 \mu\text{eV}$ selected for the data analysis. The chemicals were purchased from Alfa Aesar. Selenium shots of 99.999% purity were loaded into an aluminum sample holder of annular geometry with 1 mm gap size, where they were subsequently melted *in situ* under vacuum before the measurement at 623 K. The data used for the resolution function were subsequently collected at the baseline temperature of 15 K. Gallium of 99.999% purity was loaded in the dry air as an ambient temperature liquid into an alumina sample holder of annular geometry with 2 mm gap size. Standard aluminum sample holders could not be used to contain gallium, and proper precautions needed to be taken due to gallium hygroscopic properties. The measurement was done at 363 K under vacuum; any possible residual amounts of water were removed under *in situ* evacuation. Similar to selenium, the data used for the resolution function in the gallium experiment were subsequently collected at the baseline temperature of 15 K. Sufficient space was left in the sample holder to allow for expansion in the vertical direction upon solidification. Ultrapure glycerol with less than 0.05% of impurities was loaded in the dry air into an aluminum sample holder of annular geometry with 0.05 mm gap size. Glycerol could possibly present the most problems with multiple scattering that would negatively affect the data quality, especially in the low- Q region of interest, because of the large incoherent neutron scattering cross-section of hydrogen in excess of 80 barn. Therefore, we used the 0.05 mm gap size to ensure greater than 96% transmission of neutron beam through the sample so that the effect of multiple scattering could be minimized. Following the measurement at 360 K, the data for the resolution function were collected at the baseline temperature of 5 K.

Besides T_m and T_g , the MCT critical temperature, T_c , is another relevant parameter for glass-formers. For selenium, $T_m = 494 \text{ K}$, $T_g \approx 305 \text{ K}$ [24], $T_c = 330 \text{ K}$ [5]. For glycerol, $T_m = 291 \text{ K}$, $T_g \approx 185 \text{ K}$, $T_c \approx 288 \text{ K}$ [25]. For non glass-forming gallium, $T_m = 303 \text{ K}$. Therefore, for all three materials our measurements were performed at $T \approx 1.2T_m$.

3. Results and discussion

The QENS data were fit independently at each Q value using the following expression:

$$I(Q, E) = [x(Q)\delta(E) + (1 - x(Q))S(Q, E)] \otimes R(Q, E) + B(Q, E) \quad (1)$$

Here $\delta(E)$ is a delta function centered at zero energy transfer, $x(Q)$ represents the fraction of the elastic scattering, $B(Q, E)$ is the linear background term, $B(Q, E) = C_1(Q)E + C_2(Q)$, $R(Q, E)$ is the resolution function, and $S(Q, E)$ is the dynamic structure factor. Even

though for samples in the liquid state there should be no elastic term described by the parameter x , in practice the spectra as collected contain some elastic contribution from the sample holders and the spectrometer background. Accurate subtraction of this elastic term would require precise knowledge of the sample attenuation factors and high-statistics measurements of the background and empty sample holder intensities, whereas simple subtraction will result in negative intensities due to partial shielding by the sample holders. Alternatively, the amount of elastic scattering that needs to be subtracted from the data to leave only the QENS but no elastic signal could be determined empirically for each Q value, by means of determining the parameter x in Eq. (1) and subtracting the $x(Q)R(Q, E)$ term from the original spectra. Even though such practice is sometimes employed in treatment of data from liquids, we chose not to perform the subtraction of the elastic term and simply allowed for a freely varying elastic term in the fits, which does not affect the $S(Q, E)$ that is of interest to us. We use the following expression for the dynamic structure factor:

$$S(Q, E) = (1 - p(Q)) \frac{1}{\pi} \frac{\Gamma_1(Q)}{E^2 + \Gamma_1^2(Q)} + p(Q) \frac{1}{\pi} \frac{\Gamma_2(Q)}{E^2 + \Gamma_2^2(Q)} \quad (2)$$

The parameters $p(Q)$ and $(1 - p(Q))$ represent the relative spectral weights of the narrow Lorentzian term with HWHM of Γ_2 and the broad Lorentzian term with HWHM of Γ_1 , respectively. We will discuss the physical meaning of the parameter $p(Q)$ below. When $p(Q) = 0$ at all Q values, the two-Lorentzian model is reduced to the one-Lorentzian model. Figure 1 shows the data for selenium and gallium at a representative value of $Q = 0.575 \text{ \AA}^{-1}$ that were fit using one- and two-Lorentzian models. The difference between the two materials is obvious. While the one-Lorentzian fit is obviously inadequate in the case of selenium, for gallium the one- and two-Lorentzian fits are hardly distinguishable in quality. Likewise, the difference plot for the one-Lorentzian fit of the selenium data shows a pronounced systematic mismatch between the data and fit, which is largely eliminated in the two-Lorentzian fit. On the other hand, for the gallium data, the difference plot for the one-Lorentzian fit is already adequate, and addition of the second Lorentzian component to the fit brings little improvement. Unlike for selenium, QENS data fits for gallium do not require a second Lorentzian component. This observation is corroborated by the Q -dependence of the Lorentzian fit component broadening presented in Figure 2. For selenium, the two-Lorentzian fit yields the broadening distinctively different from that given by the one component fit. While the error bars on the broader of the two components are sizable because the width of this component becomes comparable with the dynamic range of the experiment, its Q -dependence (which plateaus at low Q values) is consistent with a spatially localized dynamic process [11,12]. This process is faster than the second, long-range diffusion process, with the broadening approaching zero at low Q values. For gallium, on the other hand, incorporation of the second component into the fit yields the qualitatively unchanged broad component and the unphysically narrow (much below the resolution limit) component that is almost indistinguishable from zero. This again points to evidence of two dynamic components in the selenium, but not gallium QENS data fits. Furthermore, the long-range diffusion coefficients obtained from fitting the Q -dependence of the component that approached zero in the low- Q limit, as shown in Figure 2, can be compared with the data available in the literature. We used the well-known jump diffusion model [26], $\Gamma(Q) = \hbar D Q^2 / (1 + D Q^2 \tau)$, where τ is the residence time between jumps, to obtain the value of the long-range translational diffusion coefficient of $0.6 \times 10^{-9} \text{ m}^2/\text{s}$ for selenium at 623 K and $2.7 \times 10^{-9} \text{ m}^2/\text{s}$ for gallium at 363 K. The diffusion coefficient of $2.3 \times 10^{-9} \text{ m}^2/\text{s}$ has been reported for gallium at this temperature [27], which supports our assertion that there is a

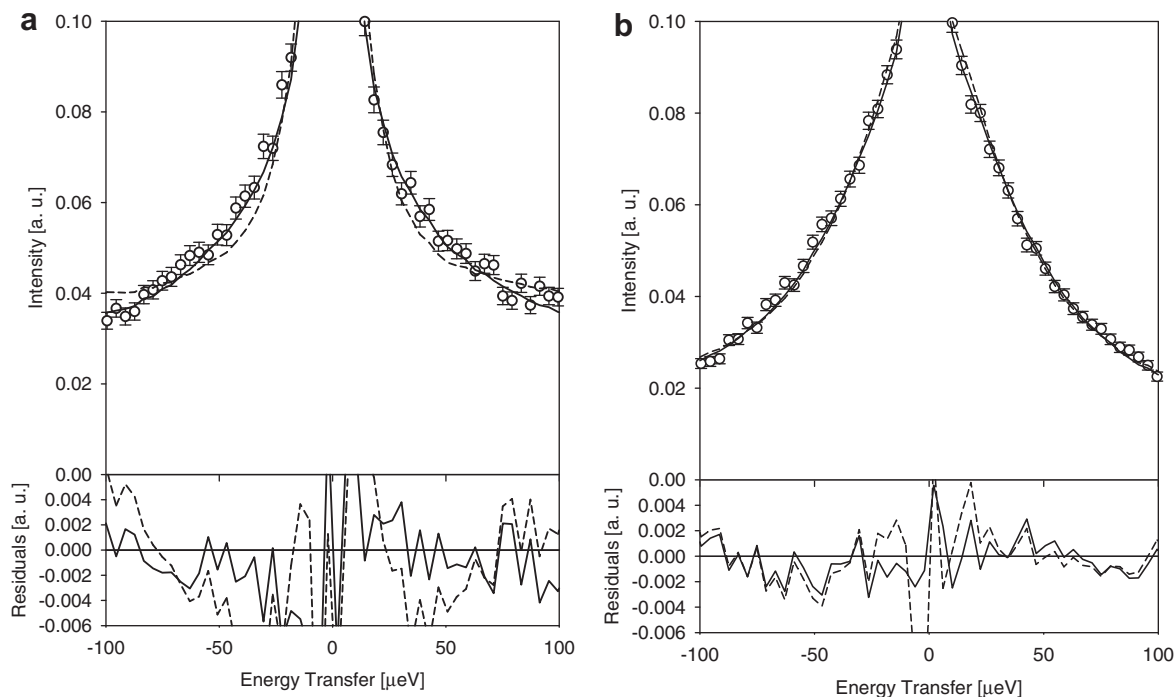


Figure 1. (a) Selenium at 623 K. (b) Gallium at 363 K. QENS data (symbols) at $Q = 0.575 \text{ \AA}^{-1}$ and fits with one-Lorentzian (dashed line) and two-Lorentzian (solid line) $S(Q, E)$ in Eq. (1). In order to facilitate comparison, the data are normalized to unity at the $S(Q, E)$ maximum and plotted using 4 μeV energy transfer bins with the elastic intensities truncated. The corresponding difference plots are shown below.

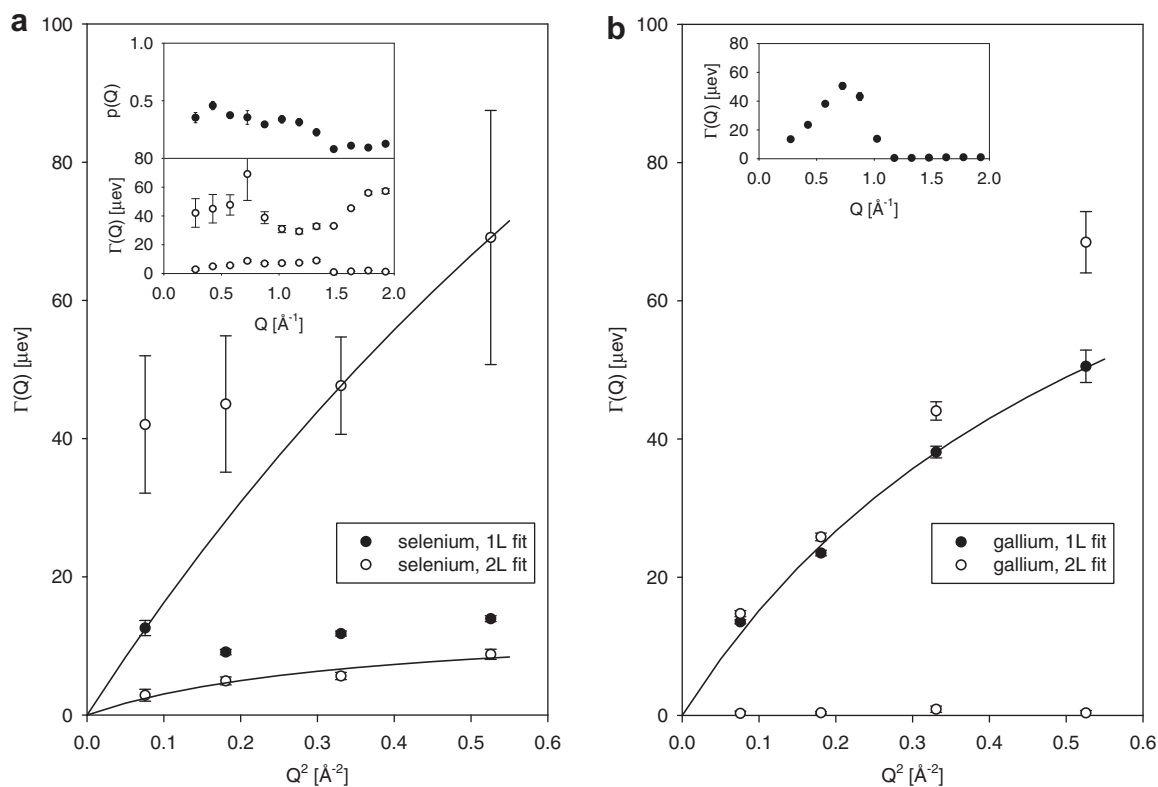


Figure 2. The HWHM of the Lorentzian components for one- (filled symbols) and two-component (open symbols) fits of the $S(Q, E)$. The Q -dependence of the HWHM from the two-component fit for selenium and the one-component fit for gallium is in turn fitted with a jump diffusion model (see the text). (a) Selenium at 623 K. Lower inset: the HWHM of the two-Lorentzian data fits over the extended Q range. Upper inset: the relative spectral weight of the narrower Lorentzian. (b) Gallium at 363 K. Inset: the HWHM of the one-Lorentzian data fits over the extended Q range.

single Lorentzian component in the QENS gallium data which represents the long-range translational diffusion. For selenium,

the values measured at 623 K [20] or extrapolated to 623 K from lower temperatures [21] are 0.6×10^{-9} and $0.55 \times 10^{-9} \text{ m}^2/\text{s}$,

respectively. Therefore, the slower of the two components in our two-Lorentzian fit represents the long-range translational diffusion in selenium. Furthermore, using the single-component fit, we obviously would not obtain the correct value of the diffusion coefficient, as one can see in Figure 2a. Thus, the presence of two dynamical components, the faster of which is spatially localized, is indeed clearly suggested by the QENS data on selenium. The spatially localized component can be described by a hybrid model that combines the features of the jump diffusion (the broadening that increases slower than Q^2 at higher Q) and localization (a plateau at lower Q).

Even though we used only the QENS broadening measured at low Q to obtain the diffusion coefficients, as shown in Figure 2, for illustrative purposes the insets in Figure 2a and b show the QENS broadening from fitting over the entire Q range of the spectrometer. The diffusion in gallium is fast enough so that the QENS signal becomes too broad for the spectrometer energy transfer range and can no longer be extracted beyond the low Q points. For selenium, the broadening remains within the accessible range, but, when approaching $Q \approx 1 \text{ \AA}^{-1}$, it enters a different regime. A crossover from hydrodynamic to atomic diffusion regime at this Q value was predicted in one MD study [28]. Although selenium is obviously not a simple liquid, our goal is to identify the common features of the spatially restricted β -relaxation processes in glass-forming liquids, regardless of the peculiarities of their atomic or molecular structure. In fact, it is the unusual structure of liquid selenium that makes this single-element liquid a glass-former [5]. In selenium, the heterogeneity in the structure plays the role in glass formation normally played by the chemical heterogeneity, even to the point that a pre-peak in the structure factor, usually indicative of short-range ordering, is also observed in liquid selenium [5]. This pre-peak at $Q \approx 1 \text{ \AA}^{-1}$ likely affects the data fits as illustrated by the insets in Figure 2a, suggesting the possible influence of the coherent scattering on the QENS signal. Even though the purely coherent scattering is an unlikely explanation for the faster component, which would still have to approach zero in the limit of low Q had it originated from the same long-range translational diffusion dynamics that yields the narrow component, possible influence of the coherent scattering on the fits cannot be discounted. For this reason, we defer the quantitative discussion of the parameters such as the relative spectral weight of the fit components to the hydrogenated material considered below, where the dominance of the incoherent scattering is unquestionable.

A Fourier-transformation of the model scattering function, $S(Q, E)$, in Eq. (2) yields the intermediate scattering function, $I(Q, t)$:

$$I(Q, t) = (1 - p(Q)) \exp[-t/(\hbar/\Gamma_1(Q))] + p(Q) \times \exp[-t/(\hbar/\Gamma_2(Q))] \quad (3)$$

On the basis of our fits of the $S(Q, E)$, there should be a single decay term in the $I(Q, t)$ for gallium and a two-step decay for selenium. The two-step decay of the $I(Q, t)$ in glass-forming liquids has been reported in numerous MD studies, including that of liquid selenium [5]. Here we would like to emphasize that the long-range translational and the faster spatially localized diffusion processes previously reported in the studies of ionic liquids [11,12] and water [13] correspond to the main α -relaxation and β -relaxation processes, respectively. Both relaxations are related to motions of the molecular center of mass. Unlike other possible secondary relaxation processes, e.g., from side groups dynamics in the more complex molecules, this β -relaxation can be appropriately referred to as the localized diffusion. This is because the molecular center of mass is not at rest, as would be the case for purely rotational motions, but rather moves in space, as evidenced by the increase with Q of the width of the fast component in QENS data [11–13] and the

Q -dependent β -relaxation times [9,10,13]. A strong argument in favor of generally non-rotational character of this β -relaxation process comes from the MD study of glass-forming metallic alloy $\text{Al}_{80}\text{Ni}_{20}$ [6], where individual atoms, unlike molecules, do not move rotationally, yet exhibit β -relaxation. Likewise, a coarse-grained visualization of MD trajectories of molecules in super-cooled water [13] shows translation-like motions of the center of mass within the boundaries of ‘basins’. The β - and α -relaxations represent the intra- and inter-‘basin’ dynamics, respectively, and both of them are of translational character, but the β -process is spatially localized.

In an archetypical molecular glass-former glycerol, the QENS signal is dominated by the incoherent scattering from the hydrogen atoms at all Q values. Nevertheless, the Q -range of the data analysis is limited by the entrance into the spectra of the localized processes not related to the center of mass motions, such, e.g., segmental motions of the skeletal CH_2 groups about the C–C bonds [29,30]. One-Lorentzian fit is so obviously inadequate for glycerol that comparison should be made not between one- and two-Lorentzian fits, but instead between the two-Lorentzian fit and the fit with Fourier-transformation of stretched exponential relaxation function, $I(Q, t) = \exp[-(t/\tau(Q))^\beta]$. The latter model is often used to describe the non-exponential relaxations in glass-forming liquids, including glycerol [31]. Because these two models yield data fits that are very similar in quality, the choice of the most appropriate model should be based on other considerations. As one can see in Figure 3, the fit results for glycerol obtained using the two-Lorentzian model are qualitatively similar in their Q -dependence to those obtained for liquid selenium. There is a broader component that tends to plateau at low Q , which originates from the spatially localized (but non-rotational) dynamic process. Similar to the broad component in selenium, the spatially localized component in glycerol combines the features of the jump diffusion (the broadening that increases slower than Q^2 at higher Q) and localization (leveling off at lower Q). The second, narrower component, originates from the slower long-range translational diffusion. For the lowest Q values, a simple Fickian diffusion law fit of this component, $\Gamma(Q) = \hbar D Q^2$ (straight line in Figure 3), yields the translational diffusion coefficient of $0.8 \times 10^{-10} \text{ m}^2/\text{s}$, in excellent agreement with the diffusion coefficient of the same value obtained from the pulsed magnetic field gradient NMR measurements of glycerol [32]. A fit of this narrow component over the entire Q range shown in Figure 3 with the jump diffusion model yields only a slightly higher value of $1.0 \times 10^{-10} \text{ m}^2/\text{s}$. Interestingly, the quantitative agreement between the QENS and NMR diffusion coefficients for glycerol has remained elusive [29–31]. In this respect, the close agreement with the NMR diffusion coefficient that we obtained using simple and physically intuitive two-Lorentzian fits is encouraging, even though one should naturally expect a better agreement between NMR and QENS data at higher temperatures, where the diffusion is faster and the resolution limitations of QENS are less significant. On the other hand, the stretched exponential model yields much less plausible Q -dependence of the inverse average relaxation times (filled symbols in Figure 3, calculated as $\hbar/\langle\tau\rangle = \hbar\beta/(\tau\Gamma(1/\beta))$, where Γ is the gamma-function and τ and β are the fit parameters in the $I(Q, t) = \exp[-(t/\tau)^\beta]$). Thus, the two-Lorentzian model better describes the center of mass dynamics in glycerol compared to the stretched exponential model. Arguments in favor of the two-Lorentzian model vs. stretched exponential model have been given before [13]; we also want to note that at sufficiently low Q , the stretched exponential model can never be applicable, since the simple Fickian diffusion behavior characterized by $\beta \rightarrow 1$ and the simple exponential decay of $I(Q, t) = \exp(-t/\tau)$ should be regained at low Q . A study of glycerol [31] puts the upper Q limit of such regime at $Q \leq 0.2\text{--}0.3 \text{ \AA}^{-1}$, and states that at higher Q values the data should be described

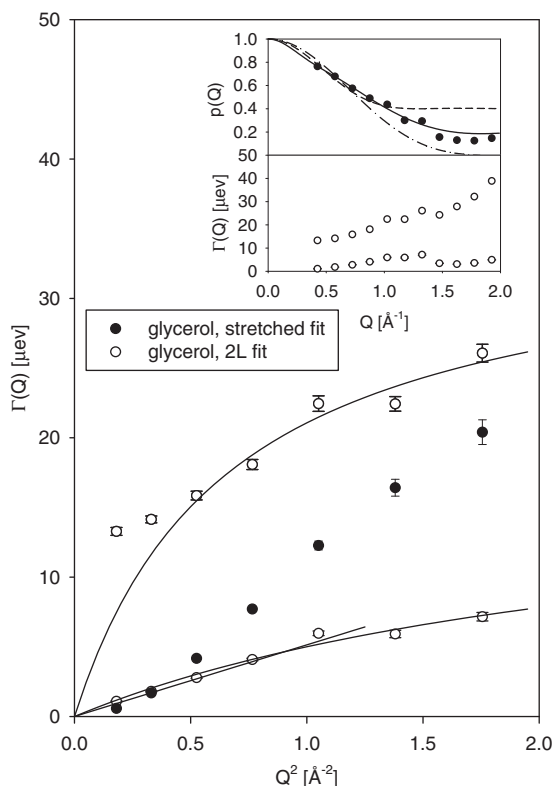


Figure 3. Glycerol at 360 K. Open symbols: HWHM of two-component Lorentzian fit. Filled symbols: inverse average relaxation times of the fits with Fourier-transformed stretched exponential (see the text). The Q -dependence of the HWHM from the two-component fit is in turn fitted with a jump diffusion model for the broad component and both Fickian diffusion and a jump diffusion model for the narrow component (see the text). Lower inset: the HWHM of the two-Lorentzian data fits over the extended Q range. Upper inset: the relative spectral weight of the narrower Lorentzian and its fits, obtained over the same Q range as shown in the main figure, with $p(Q) = (3j_1(Qa_1)/(Qa_1))^2$ (dashed-dotted line), $p(Q) = c_1 + (1 - c_1)(3j_1(Qa_1)/(Qa_1))^2$ (dashed line) and $p(Q) = [c_1 + (1 - c_1)(3j_1(Qa_1)/(Qa_1))^2][c_2 + (1 - c_2)j_0^2(Qa_2)]$ (solid line); see the text for explanation.

by the stretched exponential relaxation function with $\beta < 1$ and a complex Q -dependence of the τ and β parameters. We agree that, at low Q , only the α -relaxation, or long-range translation diffusion, can be observed. However, when at higher Q the β -relaxation becomes observable, the total $I(Q, t)$ is best described by a sum of two decays (Eq. (3)), not by a single stretched exponential term. In the energy space, this corresponds to a weighted sum of two Lorentzians (Eq. (2)), and the Q -dependence of this weight has a simple physical interpretation related to the spatial localization of the β -relaxation process ('caging' effect). This is not to say that each of the two relaxation components is best represented by a simple exponential decay. In fact, this is almost certainly an oversimplification; e.g., the β -relaxation component is characterized by the sub-diffusive time dependence of the mean-squared displacement of the center of mass inside the 'cage', $\langle x^2 \rangle(t) \sim t^\alpha$ ($\alpha < 1$) [3], and, thus, should differ from a simple exponential decay. However, in the interpretation of the QENS data it is more important to account for the clear separation of the α - and β -relaxation components then to analyze the deviations of each component from the simple exponential decay.

Importantly, for simultaneous measurements of both the slow α -relaxation and fast β -relaxation, which is necessary for determining the spectral weight, $p(Q)$, one needs a neutron spectrometer with both a high-energy resolution and a sufficiently broad dynamic range, such as the BASIS. The upper inset in Figure 3 shows fits of the $p(Q)$ obtained over the same Q range as shown

in the main figure following the approach outlined in [11]. The β -relaxation is described by $I(Q, t) = p(Q) + (1 - p(Q))\exp(-t/\tau_1(Q))$, where $p(Q)$ is the Q -dependent elastic incoherent structure factor (EISF) of the β -process localized in the 'cage'. For not too large Q values, this $I(Q, t)$ appropriately decays to a plateau level $p(Q)$ at long times. The α -relaxation is described by $I(Q, t) = \exp(-t/\tau_2(Q))$, which decays to zero at long times for all Q values. Assuming decoupling of the two processes, the resulting overall relaxation function is a product,

$$I(Q, t) = (1 - p(Q)) \exp[-t(1/\tau_1(Q) + 1/\tau_2(Q))] + p(Q) \times \exp[-t/\tau_2(Q)] \quad (4)$$

with the corresponding Fourier-transformation into the energy space:

$$S(Q, E) = (1 - p(Q)) \frac{1}{\pi} \frac{\hbar(1/\tau_1(Q) + 1/\tau_2(Q))}{E^2 + [\hbar(1/\tau_1(Q) + 1/\tau_2(Q))]^2} + p(Q) \frac{1}{\pi} \times \frac{\hbar/\tau_2(Q)}{E^2 + [\hbar/\tau_2(Q)]^2} \quad (5)$$

Eqs. (5) and (4) are equivalent to Eqs. (2) and (3), respectively, with $\tau_1 = \hbar/(\Gamma_1 - \Gamma_2)$ and $\tau_2 = \hbar/\Gamma_2$, or $\Gamma_1 = \hbar(1/\tau_1 + 1/\tau_2)$ and or $\Gamma_2 = \hbar/\tau_2$. Therefore, in the time space, the parameter $p(Q)$ represents the Q -dependent non-zero plateau value to which the β -process decays. In the energy space, the parameter $p(Q)$ represents the Q -dependent EISF of the spatially localized β -process. Most generally, the β -relaxation can be visualized as diffusion confined within an approximately spherical 'cage' or 'basin' [13] of radius a_1 , where the molecule moves between the inter-basin α -relaxation jumps. For such a confined diffusion process within a sphere, the Q -dependence of the EISF is given by $p(Q) = (3j_1(Qa_1)/(Qa_1))^2$, where j_1 is the spherical Bessel function [33]. The dashed-dotted line in the upper inset in Figure 3 shows the fit of the $p(Q)$ with this expression, which yields $a_1 = 2.4$ Å for the confinement radius. However, the quality of this fit is not satisfactory. The EISF for the localized processed can be modified to allow for a fraction of molecules, described by the parameter c_1 , that effectively do not contribute to the β -process dynamics: $p(Q) = c_1 + (1 - c_1)(3j_1(Qa_1)/(Qa_1))^2$ [11]. The parameter c_1 may be related to the presence of either static (short-range bond ordering [34,35]) or dynamic [36] heterogeneities. Such a fit shown in the inset with the dashed line yields $a_1 = 3.5$ Å and $c_1 = 0.4$; the fit quality is good up to ≈ 1 Å⁻¹, but becomes poor at higher Q values. Still better fit (solid line in the inset) could be obtained if one allows for an extra term that describes the localized dynamics of the side groups with the localization radius a_2 and non-participating fraction of hydrogen atoms c_2 : $p(Q) = [c_1 + (1 - c_1)(3j_1(Qa_1)/(Qa_1))^2][c_2 + (1 - c_2)j_0^2(Qa_2)]$ [11,12]; the fit parameters are as follows: $a_1 = 8.0$ Å, $c_1 = 0.87$, $a_2 = 1.7$ Å, $c_2 = 0.21$. Note that the value of $a_1 = 8.0$ Å is now in better agreement with the estimate of the confinement radius of 5.7–7.8 Å that, for diffusion within a sphere [33], can be computed as $3.3/Q_0$, where Q_0 is the crossover point from the translation-like Q -dependent broadening at higher Q to more plateau-like broadening at lower Q for the broader component shown in Figure 3. Even in the most generic functional form, such as EISF(Q) = $j_0^2(Qa_2)$ for diffusion on sphere with radius a_2 , introduction of the parameters a_2 and c_2 has been helpful in describing the contribution from the methyl and amino groups [11,12]. Likewise, the introduction of the term $[c_2 + (1 - c_2)j_0^2(Qa_2)]$ to describe the contribution from the segmental (not related to the center-of-mass dynamics) motions of the skeletal CH₂ groups about the C–C bonds [29,30] seems to be essential to obtain the more plausible value for the confinement radius a_1 for the fast localized dynamic component in glycerol.

We note that the $p(Q)$ should always approach unity in the limit of the low Q , where there is no contribution from the spatially localized process. The $p(Q)$ for selenium shown in the upper inset of Figure 2a seems to exhibit too low values at low Q compared to the $p(Q)$ for glycerol. Nevertheless, we have found that a suitable Q -dependence of the $p(Q)$ that approaches unity in the limit of the low Q can be obtained if we assume a cylindrical rather than spherical confinement [33] for the fast localized component in selenium. Such anisotropy could possibly be attributed to the chain-like structure of selenium [15–18].

There must be a microscopic mechanism behind the dictum that a liquid needs to have high viscosity near the melting temperature in order to be a good glass-former. The high viscosity that promotes supercooling reflects the fact that the α -relaxation (cage-breaking process related to viscosity) is well separated from the β -relaxation (fast in-cage dynamics). The lower viscosity in non glass-forming liquids reflects the lack of such a pronounced separation, even though the structural relaxation in non glass-forming liquids and α -relaxation in glass-formers may be fundamentally similar [37]. Good glass-formers exhibit such a separation even above T_m (metal alloys [38], glycerol and selenium in this work). Non glass-forming liquids show a single relaxation process not only above T_m (sodium [39], gallium in this work), but also below T_m , if they can be supercooled using special techniques (nickel [40]). We acknowledge that systematic studies of various samples would be needed to verify our hypothesis that the presence of a distinct non-rotational β -process that involves motions of the molecular center of mass in the liquid state represents a high-temperature signature of the glass-forming behavior at low temperatures.

4. Conclusion

We use QENS to compare high-temperature ($T \approx 1.2T_m$) single-particle center of mass dynamics in non glass-forming (gallium) and glass-forming (selenium and glycerol) liquids. For the glass-forming liquids, but not non glass-forming gallium, the dynamic structure factor, $S(Q, E)$, and the corresponding intermediate scattering function, $I(Q, t)$, exhibit a two-step relaxation, first through the localized β -process, then through the main α -process. The two-step relaxation, which is caused by the caging effect known in glass-forming liquids, underlines the microscopic mechanism of high viscosity that promotes supercooling and eventual glass formation. On the other hand, gallium, that can be supercooled but then spontaneously crystallizes without forming glass, exhibits no caging effect in its relaxation. While the α -relaxation represents the long-range translational diffusion, the β -relaxation is due to a process that is spatially localized, in ‘cage’ or ‘basin’, but, nevertheless involves motions of the center of mass. We demonstrate that data analysis that explicitly uses two separate components to represent the α - and β -processes yields the correct values of the long-range translational diffusion coefficient determined by the α -relaxation. In the framework of the confined diffusion model, the ratio of the relative spectral weights of the two components can be used to experimentally determine the ‘basin’ size, which cannot be obtained by other techniques except from MD simulations.

Note added in proof

The evidence of the spatially confined center of mass relaxation component in glycerol was obtained in this study at 360 K. Perhaps, a better illustration of the fact that this temperature point represents truly high-temperature regime for glycerol comes from the parabolic fit of the temperature dependence of the glycerol

structural relaxation times [41]. In this fit, the temperature of 338 K was identified as T_0 , the temperature above which correlated dynamics is not required for molecular motions, and temperature variation of transport is nearly negligible. In other words, the super-Arrhenius behavior of the relaxation time, characteristic of glass-formers, is not observed above T_0 . Remarkably, even well above the $T_0 = 338$ K, QENS data from glycerol already show the presence of the localized relaxation center of mass dynamics, a precursor of eventual glass formation at much lower temperatures.

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